GGG-B-481a
FEBRUARY 16, 1954
SUPERSEDING
GGG-B-00481 (Navy-Ships)
3 November 1952

FEDERAL SPECIFICATION

BLOCK, DOLLY, SHEET METAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers dolly blocks for use in straightening, swaging, and finishing by hand, sheet metal bodies and fenders of automobiles and other motor vehicles.

1.2 **Classification.**—Dolly blocks shall be of the following types and styles, as specified (see 6.1):

Type I.—General purpose.
  Style A.—High crown.
  Style B.—Low crown.
  Style C.—Anvil.
  Style D.—Rail.
Type II.—Bead.
Type III.—Heel.
Type IV.—Toe.
Type V.—Wedge.
Type VI.—Body.
Type VII.—Finger.
Type VIII.—Shrinking.
  Style A.—With trenching grooves.
  Style B.—Without trenching grooves.
Type IX.—Fender.
Type X.—Angle.
Type XI.—Roughing.
Type XII.—Thin high crown.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

NN-B-591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601—Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621—Boxes; Wood, Nailed and Lock-Corner.
NN-B-631—Boxes; Wood, Wirebound (for Domestic Shipment).
LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636—Boxes; Fiber, Solid (for Domestic Shipment).

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Federal Specifications and the Index of Federal Specifications and Standards may be obtained upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 8, Seventh and D Streets SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices. Single copies of this specification and other product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kan-

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sas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Copies of the Military Standard referenced above required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Specifications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 **Material.**—Materials used in the construction of dolly blocks shall be as hereinafter specified for each type and style. All materials shall be of uniform quality, free of porosity, hard spots, cracks or other injurious defects. Strength and other essential physical properties shall be adequate throughout for the purpose intended.

3.2 **Marking for identification.** — Dolly blocks shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined. No engraving, stamping, or other symbol markings to identify the brand shall be made on the working surfaces of the dolly blocks.

3.3 **Finish.**

3.3.1 *Surface.*—All surfaces of the dolly blocks shall be finished in a workmanlike manner consistent with the functional requirements of the dolly blocks. All radii and contours shall be blended, and shall be free of burrs and sharp edges. Surfaces usually ground, buffed, or otherwise finished shall have a surface finish equivalent to the finish produced by 80-grit or finer abrasive.

3.3.2 *Coating.*—The dolly blocks shall have one of the following coatings:

(*a*) Chemically produced oxide coating, oil treated.

(*b*) Chemically produced phosphate coating, oil treated.

(*c*) Lacquer.

3.4 **Physical requirements.**—All types and styles of dolly blocks shall conform to table I, as applicable.

TABLE I.—*Dimensions for dolly blocks*

Type	Style	Length		Width		Thickness		Weight (approx.)
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
		Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*
I	A	2 5/16	3 1/4	2 5/16	2 9/16	1 3/8	2 1/8	2 3/4
	B	3 1/16	3 1/4	2 3/8	2 5/8	1 13/16	2 1/8	2 3/4
	C	3 13/16	4 1/8	2 11/16	2 13/16	2 1/16	2 1/8	3
	D	2 11/16	3 1/8	2 1/4	2 7/8	2 5/16	2 7/16	2 3/4
II		4 3/8	4 1/2	2 1/2	2 5/8	1 13/16	1 15/16	2 1/2
III		8	8 1/8	2 5/16	2 7/16	1 1/16	1 1/2	2 1/2
IV		4 1/2	4 13/16	2 5/16	2 13/16	7/8	1 3/8	2 1/2
V		5 1/16	5 13/16	2 1/8	2 5/8	1 1/16	1 1/16	3 1/2
VI		4 7/16	4 5/8	2 5/16	2 3/8	1 1/2	1 13/16	3 1/4
VII		3 5/8	3 11/16	2 5/8	2 3/4	2 1/16	2 1/8	2 1/4
VIII	A	2 15/16	3 1/16	2 1/2	2 5/8	1 13/16	1 3/4	2 1/2
	B	3 1/4	3 3/8	3	3 1/8	1 1/16	1 1/8	2 1/2
IX		3 5/8	3 13/16	2 11/16	3 1/4	1 13/16	1 15/16	2 1/4
X		3 11/16	3 13/16	2 5/16	2 3/8	1 1/8	1 1/16	2
XI		4	4 3/8	4 1/16	4 3/8	2 11/16	2 13/16	5 1/2
XII		4	4 3/8	2 1/2	2 3/4	1	1 1/4	2 1/4

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3.5 Type I, general purpose.—General-purpose blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale.

3.5.1 *Style A, high crown.*—High-crown blocks shall provide suitable working surfaces for use on high-crown fenders, panels, and hoods, and shall be similar to figure 1.

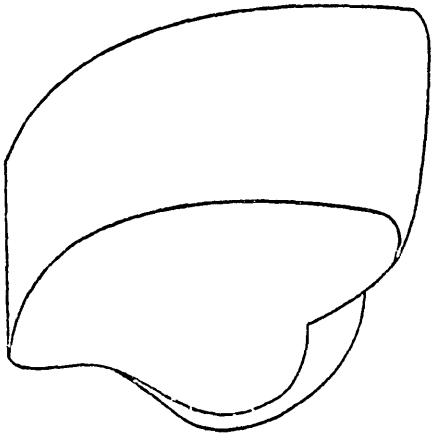


FIGURE 1.—Type I, style A, general purpose, high crown.

3.5.2 *Style B, low crown.* — Low-crown blocks shall provide suitable working surfaces for use on low-crown panels and flat portions of the body, and shall be similar to figure 2.

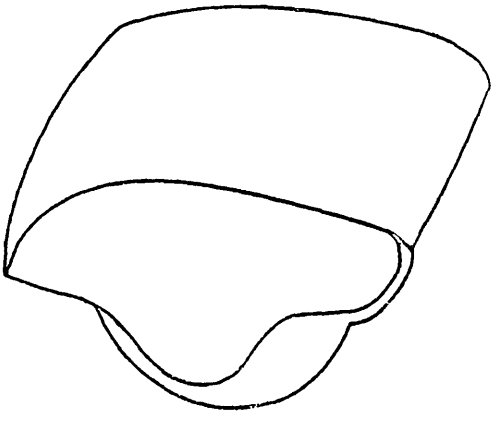


FIGURE 2.—Type I, style B, general purpose, low crown.

3.5.3 *Style C, anvil.* — Anvil blocks shall provide working surfaces suitable for various curves and contours of the body, door panels, and deep-skirted fenders, and shall be similar to figure 3.

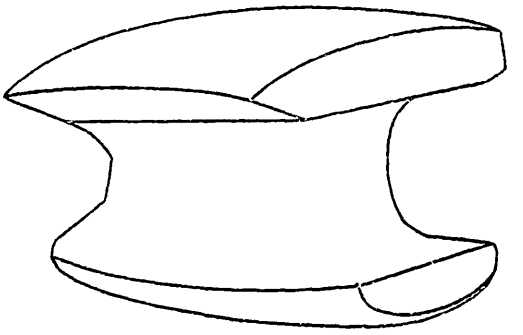


FIGURE 3.—Type I, style C, general purpose, anvil.

3.5.4 *Style D, rail.*—Rail blocks shall be suitable for bumping, dinging, straightening, and finishing various portions of the body, door panels, and fenders. The block shall also be provided with two beading and flanging lips, and shall be similar to figure 4.

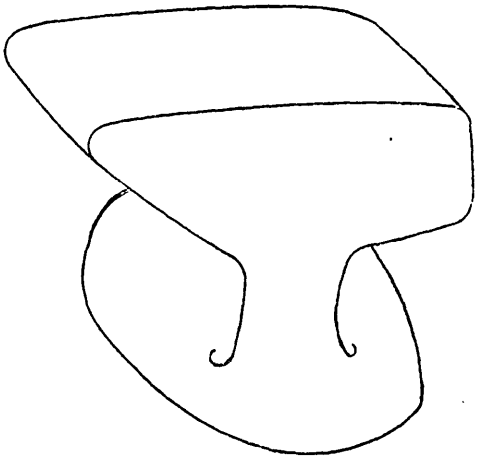


FIGURE 4.—Type I, style D, general purpose, rail.

3.6 Type II, bead.—Bead blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the

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C scale, and shall be similar to figure 5. The working surfaces shall be suitable for use on long curved fenders, beads and flanges.

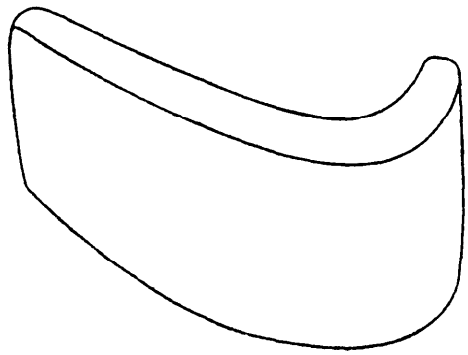


FIGURE 5.—Type II, bead dolly block.

3.7 **Type III, heel.**—Heel blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale, and shall be similar to figure 6. The working surface shall be adaptable for use in sharp corners and wide radii portions of body and panels.

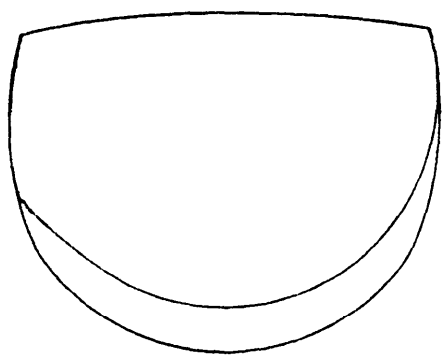


FIGURE 6.—Type III, heel dolly block.

3.8 **Type IV, toe.**—Toe blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale, and shall be similar to figure 7. The block shall be provided with a flat face and flat edge for use in dinging flat surfaces and edges of the lower portion of the hood and other flat panels. The crowned and curved portion of the block shall be suitable for use in odd crowns and contours of the body and panels.

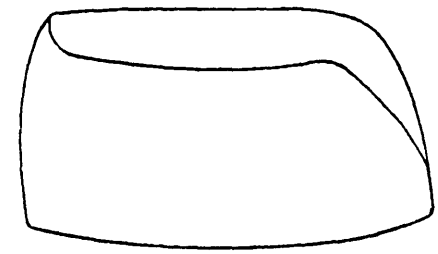


FIGURE 7.—Type IV, toe dolly block.

3.9 **Type V, wedge.**—Wedge blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale, and shall be similar to figure 8. The shape of the block shall be suitable for use behind brackets, braces, and other reinforcements of the roof, body, fenders, and panels.

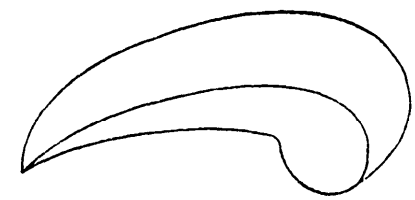


FIGURE 8.—Type V, wedge dolly block.

3.10 **Type VI, body.**—Body blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale, and shall be similar to figure 9. The crowns and radii shall be suitable for use on various curves and contours of the body.

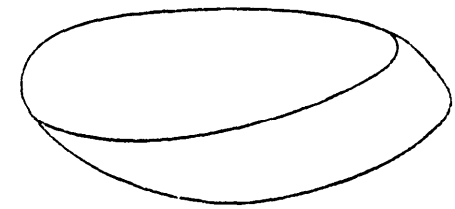


FIGURE 9.—Type VI, body dolly block.

3.11 **Type VII, finger.**—Finger blocks shall be of steel or malleable iron having a Rockwell hardness of not less than 40 nor more than 58 on the C scale for steel and not less than 20 nor more than 25 on the C scale for malleable iron. The block shall be similar to figure 10, and shall be adaptable for bump hammering on portions of the body where space is limited due to inner construction and adjoining metals.

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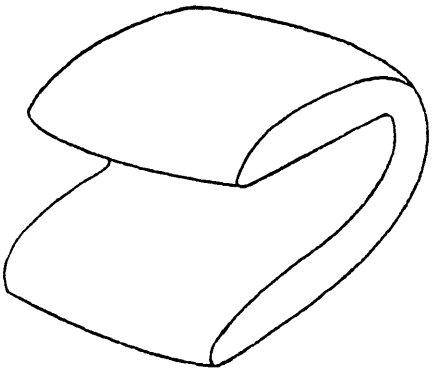


FIGURE 10.—Type VII, finger dolly block.

3.12 **Type VIII, shrinking.** — Shrinking blocks shall be of steel or malleable iron. The blocks shall have a Rockwell hardness of not less than 40 nor more than 58 on the C scale for steel and not less than 20 nor more than 25 on the C scale for malleable iron.

3.12.1 *Style A, with trenching grooves.*— The block shall be provided with high and low crowns suitable for hot shrinking, raising beads, and molding, and shall be similar to figure 11. Wide and narrow trenching grooves shall be provided for setting in welded seams for paddle soldering.

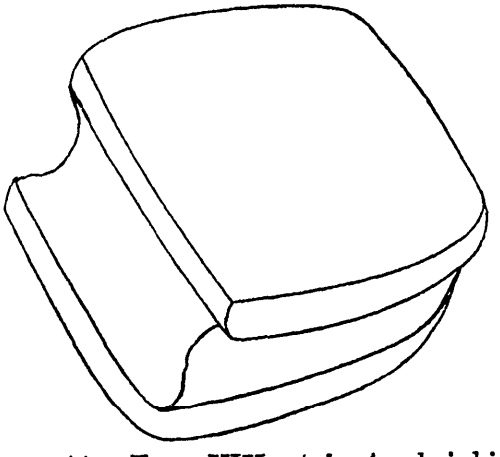


FIGURE 11.—Type VIII, style A, shrinking, with trenching grooves.

3.12.2 *Style B, without trenching grooves.*— The block shall be provided with a low crown suitable for shrinking body metal, raising beads and molding, and shall be similar to figure 12.

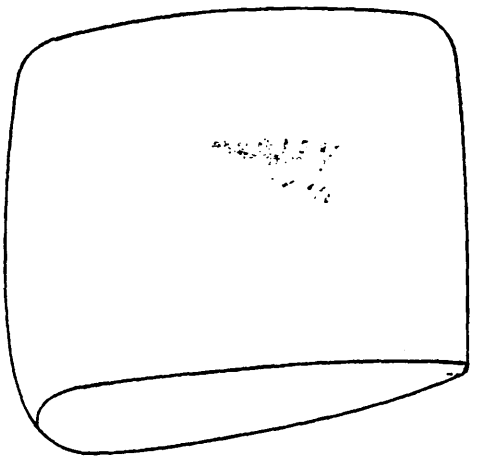


FIGURE 12.—Type VIII, style B, shrinking, without trenching grooves.

3.13 **Type IX, fender.**—Fender blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale, and shall be similar to figure 13. The radii and contours of the block shall be suitable for rolled edges and fender beads.

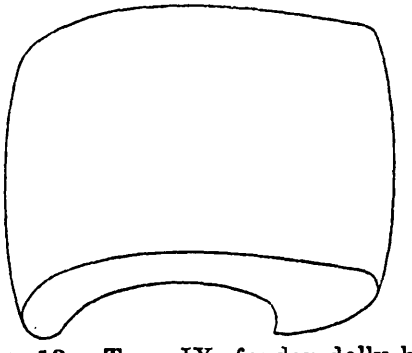


FIGURE 13.—Type IX, fender dolly block.

3.14 **Type X, angle.**—Angle blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale. The blocks shall be suitable for use on various angles, creases and curves in the body panels and fenders, and shall be similar to figure 14.

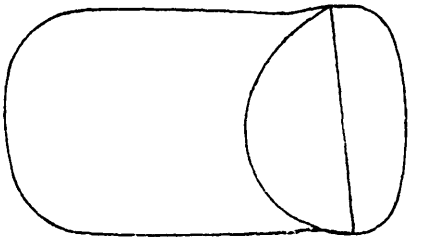


FIGURE 14.—Type X, angle dolly block.

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3.15 Type XI, roughing.—Roughing blocks shall be of forged steel having a Rockwell hardness of not less than 40 nor more than 58 on the C scale, and shall be similar to figure 15. The outside contour and side surfaces shall be suitable for roughing out deep and shallow dents with a lip provided at one end of the working surface for reshaping turned-under flanges.

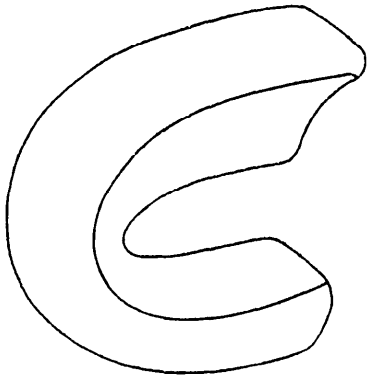


FIGURE 15.—Type XI, roughing dolly block.

3.16 Type XII, thin high crown. — Thin high crown blocks shall be of forged steel having a Rockwell hardness of not less than 45 nor more than 58 on the Rockwell C scale and shall be similar to figure 16. The blocks shall be adaptable for work where space is limited due to inner construction and adjoining metals.

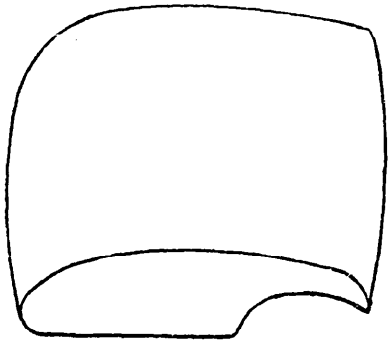


FIGURE 16.—Type XII, thin high crown dolly block.

3.17 Workmanship. — Workmanship shall be first class in every respect and the blocks shall be free from rust or any imperfections which may affect their appearance, durability, and serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All dolly blocks of the same type and style presented at one time shall be considered a lot for purpose of acceptance, inspection, and tests.

4.1.2 *Sampling for inspection.*—A random sample of dolly blocks shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with the procedures of Military Standard MIL-STD-105.

TABLE II.—*Sampling for visual and dimensional inspection AQL (approx.) = 2.5 percent defective*

Number of dolly blocks in an inspection lot	Number of dolly blocks in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	1	2
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1,300	75	4	5
1,301 and over	110	6	7

NOTE: The Government inspector may institute tightened or reduced inspection in accordance with Section 9 of Military Standard MIL-STD-105.

4.1.3 *Sampling for lot-acceptance test.*—A random sample of dolly blocks shall be selected from each inspection lot in accordance with table III and subjected to the hardness test specified in 4.3.2.1.

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TABLE III.—*Sampling for lot acceptance test AQL (approx.) = 1.0 percent defective*

Number of dolly blocks in an inspection lot	Number of dolly blocks in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	0	1
301 to 500	25	0	1
501 to 800	35	1	2
801 to 1,300	50	1	2
1,301 and over	75	2	3

4.2 **Inspection.**

4.2.1 *Visual and dimensional inspection.*—Each of the sample dolly blocks selected in accordance with table II shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any dolly block in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective dolly blocks in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 **Lot-acceptance test.**

4.3.1 *Testing of sample dolly blocks for lot acceptance.*—Each of the sample dolly blocks selected in accordance with table III shall be tested for hardness in accordance with 4.3.2.1 to verify compliance with this specification. Any sample dolly block which does not conform to the hardness requirements shall be rejected, and if the number of nonconforming dolly blocks in any sample exceeds the acceptance number for that sample the lot represented by the sample shall be rejected.

4.3.2 *Test procedures.*

4.3.2.1 *Hardness test.*—Each of the sample dolly blocks selected in accordance with 4.1.3 shall be tested to determine whether the dolly blocks conform to the required Rockwell hardness. All hardness readings shall be made after removing the outer surface to a depth of 1/64 inch.

4.4 **Rejected lots.**—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming dolly blocks. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with the specification requirements.

5. PREPARATION FOR DELIVERY

5.1 **Cleaning and preservation.**—Blocks shall be cleaned and preserved in accordance with standard commercial practice.

5.2 **Packaging.**—Unless otherwise specified, commercial packages are acceptable, provided that only one type and style is contained in any one box or package.

5.3 **Packing.**—Blocks, packaged as specified in 5.2, shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes conforming to Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, or LLL–B–636, respectively. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

5.4 **Marking.**

5.4.1 *Interior containers.*—Each box or package shall be plainly marked with the name of the article, the type, the style, and the quantity contained therein, as defined by the contract or order under which shipment is made, and the name of the manufacturer.

5.4.2 *Exterior containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the article, the type, the style, the quantity contained therein, as defined by the contract or order

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under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 **Ordering data.**—Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style of dolly block required. (See 1.2.)
(c) Whether preparation for domestic or overseas shipment is required. (See sec. 5 and 7.2.2.)

Purchasers should also exercise any desired options offered herein (see 3.3.2, 5.2 and 5.4.2).

6.2 THE ILLUSTRATIONS HEREIN ARE DESCRIPTIVE AND NOT RESTRICTIVE, and are not intended to preclude the purchase of dolly blocks otherwise conforming to this specification.

6.3 **Transportation description.** — The proper transportation description applicable to these items is:

Tools, not otherwise indexed by name.
Carload weight 30,000 pounds.
Mechanics' hand tools, not otherwise indexed.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification, of the issue in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army, Navy, and Air Force.**

7.2.1 *Applicable specifications.*

Military Specification:

MIL–P–15424—Packaging of Hand Tools; General Specification for Domestic and Overseas Preservation.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Preparation for delivery.* — Blocks shall be cleaned, preserved, packaged, packed, and marked in accordance with Military Specification MIL–P–15424. The nomenclature shall be as follows:

Block, Dolly, Sheet Metal

MILITARY INTEREST:
Army—E O
Navy—Sh A MC S Y
Air Force.

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